

Chapter 21: Migration Case Studies

Learning Objectives

By the end of this chapter, you will:

- Understand common enterprise migration patterns involving cloud, data, and AI modernization.
- Learn how AI TPMs manage large-scale migration programs across multiple teams and stakeholders.
- Apply lessons learned from real-world migration scenarios to future AI and cloud initiatives.

Beginner Explanation

Migration means moving something from one place to another.

Examples include:

- Moving data from Teradata to Azure.
- Replacing on-premises systems with cloud services.
- Modernizing ETL platforms.
- Migrating legacy applications to AI-enabled solutions.

Think of migration as moving houses.

You need to answer questions such as:

- What are we moving?
- What should be moved first?
- What are the risks?
- How do we avoid downtime?

Enterprise migrations are similar, except they involve millions of records, hundreds of applications, and thousands of users.

Typical Migration Types

Migration Type	Example
Data Migration	Teradata → Azure
Application Migration	On-Prem → Cloud
Platform Migration	Legacy ETL → Databricks
AI Modernization	Manual Processes → AI Agents

For AI TPMs, migration programs are among the most visible and complex initiatives in an organization.

Intermediate Explanation

Migration programs generally follow a standard lifecycle.

Migration Lifecycle

1. Assessment
2. Planning
3. Pilot
4. Execution
5. Validation
6. Cutover
7. Hypercare
8. Closure

Migration Workstreams

Workstream	Responsibility
Engineering	Build solutions
Infrastructure	Provision environments
Security	Conduct reviews
Data Teams	Validate data
QA Teams	Testing
TPM Office	Program governance

Success Metrics

Metric	Target
Downtime	Minimal
Data Accuracy	>99.9%
Budget Variance	<5%
Delivery Timeline	On Schedule
User Satisfaction	>90%

Common Challenges

- Scope changes
 - Data quality issues
 - Stakeholder alignment
 - Cutover risks
 - Resource constraints
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Why This Matters for an AI TPM

Responsibility	Impact
Delivery	Coordinates complex migration activities across teams.
Risk	Identifies dependencies and mitigates cutover risks.
Stakeholder Management	Maintains alignment throughout the migration lifecycle.
Budget	Ensures migrations remain within approved budgets.

Enterprise Example 1: Teradata to Azure Migration

Business Problem

A financial institution operated a legacy Teradata platform that was:

- Expensive
- Difficult to scale
- Limited in analytics capabilities

Solution

Migrate workloads to Azure Databricks and Azure Data Lake.

Architecture

- Teradata
- Azure Data Lake
- Azure Databricks
- Azure DevOps
- Power BI

Outcome

Metric	Result
Cost Reduction	28%
Query Performance	Improved by 45%
Data Availability	99.9%
Delivery Success	Achieved

Lessons Learned

- Stakeholder communication is critical.
 - Parallel runs reduce migration risks.
 - Data validation should be automated.
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Enterprise Example 2: Legacy ETL to Databricks

Business Problem

An organization maintained more than 1,500 ETL jobs using legacy tools.

Solution

Modernize workloads using:

- Azure Databricks
- Airflow
- Azure DevOps

Outcome

Metric	Result
ETL Runtime Reduction	50%
Operational Costs	Reduced by 22%
Developer Productivity	Increased significantly

Lessons Learned

- Automation accelerates migrations.
 - CI/CD reduces deployment risks.
 - Hypercare planning is essential.
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Enterprise Example 3: AI Transformation

Business Problem

A global enterprise wanted to replace manual support processes with AI capabilities.

Solution

Deploy:

- Azure OpenAI
- Azure AI Search
- AI Agents
- Microsoft Teams Integration

Outcome

Metric	Result
Ticket Reduction	55%

Metric	Result
Employee Satisfaction	91%
ROI	Achieved within 9 months

Lessons Learned

- AI adoption requires change management.
- Governance should begin early.
- User training improves adoption.

Azure Implementation

Services Used

Service	Purpose
Azure OpenAI	AI modernization
Azure AI Search	Enterprise knowledge retrieval
Azure Functions	Workflow automation
Azure AI Foundry	AI evaluations and governance

Supporting Services

- Azure Databricks
- Azure Data Lake
- Azure DevOps
- Azure Monitor
- Power BI

Migration Workflow

Step 1: Assess Current State

Document:

- Applications
- Dependencies
- Risks
- Costs

Step 2: Plan Migration

Define:

- Scope
 - Timeline
 - Resources
 - Success metrics
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Step 3: Execute Pilot

Validate:

- Performance
 - Costs
 - User experience
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Step 4: Conduct Cutover

Activities:

- Data migration
 - Validation
 - Monitoring
 - Communication
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Step 5: Hypercare

Track:

- Incidents
 - Performance
 - Adoption
 - User feedback
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Mermaid Diagram

```
```mermaid id="kek3an" graph LR
```

```
A[Assessment]
```

```
A --> B[Planning]
```

```
B --> C[Pilot]
```

C --> D[Migration]

D --> E[Validation]

E --> F[Cutover]

F --> G[Hypercare]

G --> H[Closure]

```

Enterprise Migration Architecture

```mermaid id="b0efxv"
graph TD
    A[Legacy Systems]
    A --> B[Azure Data Lake]
    B --> C[Azure Databricks]
    C --> D[Azure OpenAI]
    D --> E[Power BI]
    E --> F[Business Users]
```

TPM Checklist

- ☐ Complete migration assessment.
 - ☐ Identify dependencies.
 - ☐ Create migration roadmap.
 - ☐ Define success metrics.
 - ☐ Conduct pilot testing.
 - ☐ Establish hypercare plans.
 - ☐ Configure monitoring.
 - ☐ Validate data quality.
 - ☐ Conduct stakeholder reviews.
 - ☐ Capture lessons learned.
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Templates

Migration Dashboard

KPI	Status
Schedule	Green
Budget	Green
Risks	Yellow
Data Validation	Green
Adoption	Green

Migration Risk Register

Risk	Impact	Mitigation
Data Loss	High	Backup strategy
Downtime	High	Parallel run
Budget Overruns	Medium	Weekly reviews
Low Adoption	Medium	Training programs

Cutover Checklist

Task	Status
Backup Completed	Yes/No
Validation Completed	Yes/No
Monitoring Enabled	Yes/No
Stakeholders Notified	Yes/No
Rollback Plan Prepared	Yes/No

Common Mistakes

1. Underestimating dependencies.
2. Skipping pilot migrations.
3. Ignoring change management.
4. Failing to create rollback plans.
5. Neglecting hypercare activities.
6. Poor stakeholder communication.

7. Inadequate data validation.

Key Takeaways

- Migration programs are among the most complex initiatives managed by AI TPMs.
 - Successful migrations require planning, governance, and strong stakeholder management.
 - Pilot programs and parallel runs significantly reduce risks.
 - Hypercare is an essential phase often overlooked by organizations.
 - Azure provides a comprehensive platform for cloud and AI modernization initiatives.
 - AI TPMs should treat every migration as both a technical and organizational transformation.
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